

## CORPUS CHRISTI NAS, TX, USA

WMO: 722515

Lat: 27.683N

Lon: 97.283W

Elev: 18

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12926

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 37.1       | 41.0 | 23.0                          | 17.3 | 47.2 | 27.5 | 21.5 | 49.3 | 30.3                  | 53.3 | 27.4 | 50.7 | 15.4                  | 0    | 0.538 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.1                   | 92.8            | 80.0 | 91.6 | 80.0 | 90.8 | 79.8 | 82.8                | 88.6 | 82.1 | 88.0 | 81.5 | 87.2 | 14.5                 | 150  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 81.4                            | 163.1 | 85.0 | 80.7 | 159.3 | 84.9 | 79.7 | 154.5 | 84.8 | 46.6          | 88.2 | 45.7 | 88.6 | 45.0 | 87.7 | 87.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 26.7              | 24.4 | 21.7 | DB                         | 32.4 | 96.7               | 4.3 | 3.9                                                | 29.3 | 99.5       | 26.7 | 101.8      | 24.3 | 104.0      | 21.2 | 106.8 |
|                   |      |      | WB                         | 28.9 | 84.0               | 3.6 | 1.6                                                | 26.2 | 85.1       | 24.1 | 86.1       | 22.1 | 86.9       | 19.4 | 88.1  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 73.4   | 58.6  | 61.9  | 67.0  | 73.3  | 79.1  | 83.8  | 84.7  | 85.2  | 82.2  | 76.3  | 67.7  | 60.4  |
|                                                                        | DBStd       | 11.31  | 8.63  | 8.50  | 7.19  | 5.16  | 3.79  | 2.43  | 1.92  | 2.17  | 3.84  | 6.38  | 8.21  | 8.61  |
|                                                                        | HDD50       | 63     | 28    | 14    | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 15    |
|                                                                        | HDD65       | 714    | 229   | 141   | 65    | 9     | 0     | 0     | 0     | 0     | 0     | 7     | 67    | 196   |
|                                                                        | CDD50       | 8610   | 296   | 347   | 529   | 700   | 902   | 1013  | 1076  | 1092  | 967   | 816   | 535   | 337   |
|                                                                        | CDD65       | 3786   | 32    | 55    | 126   | 259   | 437   | 563   | 611   | 627   | 517   | 358   | 149   | 52    |
|                                                                        | CDH74       | 38186  | 48    | 105   | 337   | 1286  | 3872  | 6777  | 7672  | 8007  | 5959  | 3199  | 797   | 127   |
|                                                                        | CDH80       | 14406  | 1     | 7     | 27    | 196   | 982   | 2744  | 3370  | 3708  | 2376  | 885   | 105   | 5     |
| (14) Wind                                                              | WSAvg       | 12.1   | 11.8  | 12.5  | 13.1  | 14.0  | 13.9  | 12.6  | 12.2  | 11.0  | 10.1  | 11.2  | 11.6  | 11.6  |
| (15) Precipitation                                                     | PrecAvg     | 31.4   | 1.7   | 1.9   | 1.3   | 2.0   | 3.3   | 2.6   | 2.2   | 2.7   | 5.5   | 4.8   | 2.3   | 1.6   |
|                                                                        | PrecMax     | 59.2   | 5.3   | 5.9   | 4.0   | 9.4   | 15.7  | 9.0   | 12.2  | 8.7   | 20.7  | 21.8  | 9.8   | 7.0   |
|                                                                        | PrecMin     | 15.4   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.4   | 0.2   | 0.0   | 0.1   |
|                                                                        | PrecStd     | 11.5   | 1.3   | 1.6   | 1.2   | 2.3   | 3.1   | 2.3   | 2.6   | 2.3   | 4.3   | 4.8   | 2.2   | 1.7   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 77.8  | 80.2  | 82.7  | 88.5  | 90.6  | 93.5  | 93.2  | 95.0  | 94.5  | 90.2  | 85.7  | 80.7  |
|                                                                        |             | MCWB   | 67.6  | 70.0  | 71.4  | 69.5  | 78.8  | 80.8  | 80.3  | 79.7  | 78.0  | 79.7  | 76.4  | 72.3  |
|                                                                        | 2%          | DB     | 75.1  | 77.1  | 79.8  | 84.1  | 88.2  | 91.2  | 91.9  | 93.0  | 91.7  | 88.4  | 82.7  | 77.5  |
|                                                                        |             | MCWB   | 67.9  | 69.6  | 71.0  | 73.1  | 78.0  | 80.0  | 80.1  | 80.3  | 79.8  | 79.0  | 74.5  | 70.9  |
|                                                                        | 5%          | DB     | 72.9  | 74.7  | 77.8  | 81.9  | 86.4  | 90.1  | 91.0  | 91.8  | 90.4  | 86.6  | 80.9  | 75.1  |
|                                                                        |             | MCWB   | 66.2  | 67.8  | 69.9  | 73.5  | 77.4  | 79.7  | 80.0  | 80.1  | 79.5  | 77.5  | 73.8  | 69.6  |
|                                                                        | 10%         | DB     | 70.5  | 72.4  | 75.4  | 80.3  | 84.8  | 89.1  | 90.2  | 90.9  | 89.1  | 84.5  | 78.9  | 72.9  |
|                                                                        |             | MCWB   | 65.7  | 66.9  | 68.7  | 72.7  | 76.8  | 79.4  | 79.9  | 79.9  | 78.8  | 76.3  | 72.5  | 68.7  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 71.8  | 73.4  | 74.9  | 78.4  | 82.3  | 83.4  | 83.1  | 83.9  | 83.6  | 82.2  | 79.3  | 74.4  |
|                                                                        |             | MCDB   | 74.7  | 77.4  | 79.9  | 82.8  | 85.4  | 90.0  | 89.7  | 90.4  | 89.3  | 86.4  | 82.5  | 78.3  |
|                                                                        | 2%          | WB     | 70.3  | 71.4  | 73.2  | 76.6  | 80.4  | 82.2  | 82.2  | 82.7  | 82.2  | 80.6  | 77.0  | 72.9  |
|                                                                        |             | MCDB   | 72.7  | 74.6  | 77.4  | 80.9  | 85.0  | 88.2  | 88.5  | 88.7  | 87.9  | 85.3  | 80.8  | 76.1  |
|                                                                        | 5%          | WB     | 68.9  | 70.1  | 72.1  | 75.4  | 79.3  | 81.4  | 81.6  | 82.0  | 81.2  | 79.5  | 75.3  | 71.4  |
|                                                                        |             | MCDB   | 71.3  | 72.9  | 75.6  | 79.6  | 84.3  | 87.0  | 87.6  | 88.1  | 87.0  | 84.5  | 79.2  | 73.8  |
|                                                                        | 10%         | WB     | 67.1  | 68.9  | 71.0  | 74.5  | 78.3  | 80.6  | 81.0  | 81.3  | 80.3  | 78.3  | 73.9  | 69.8  |
|                                                                        |             | MCDB   | 69.9  | 71.3  | 73.9  | 78.4  | 83.0  | 86.3  | 87.0  | 87.4  | 86.1  | 83.3  | 77.5  | 72.3  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 14.9  | 13.9  | 13.1  | 12.3  | 10.8  | 10.9  | 11.1  | 12.1  | 12.8  | 13.2  | 14.0  | 14.5  |
|                                                                        |             | MCDBR  | 16.4  | 14.9  | 13.1  | 12.9  | 11.7  | 11.4  | 11.8  | 12.6  | 13.5  | 12.1  | 13.6  | 15.2  |
|                                                                        |             | MCWBR  | 11.0  | 9.0   | 7.3   | 6.7   | 5.1   | 4.3   | 4.1   | 4.2   | 5.5   | 6.2   | 8.1   | 10.5  |
|                                                                        | 5% WB       | MCDBR  | 13.4  | 12.5  | 11.3  | 10.6  | 10.1  | 10.2  | 10.6  | 11.6  | 12.1  | 11.3  | 12.5  | 13.6  |
|                                                                        |             | MCWBR  | 10.3  | 8.7   | 7.1   | 6.2   | 5.5   | 4.7   | 4.3   | 4.4   | 6.0   | 7.0   | 8.8   | 10.9  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.363 | 0.369 | 0.388 | 0.430 | 0.470 | 0.447 | 0.463 | 0.451 | 0.439 | 0.393 | 0.378 | 0.353 |
|                                                                        | taud        |        | 2.473 | 2.459 | 2.400 | 2.288 | 2.210 | 2.336 | 2.309 | 2.341 | 2.335 | 2.462 | 2.455 | 2.515 |
|                                                                        | Ebn at Noon |        | 279   | 287   | 287   | 276   | 264   | 268   | 264   | 267   | 267   | 275   | 271   | 277   |
|                                                                        | Edh at Noon |        | 30    | 33    | 37    | 43    | 46    | 41    | 42    | 40    | 39    | 33    | 31    | 28    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 953   | 1132  | 1444  | 1722  | 1927  | 2123  | 2094  | 2002  | 1653  | 1443  | 1081  | 856   |
|                                                                        | RadStd      |        | 103   | 187   | 156   | 164   | 134   | 106   | 169   | 109   | 139   | 147   | 112   | 115   |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (28 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +209  | +185  |

Nomenclature: See separate page